

15.1 Ideal Gas Law

Question Paper

Course	CIE A Level Physics
Section	15. Ideal Gases
Topic	15.1 Ideal Gas Law
Difficulty	Easy

Time allowed: 40

Score: /28

Percentage: /100

Question 1a

Define the mole.

[2 marks]

Question 1bIt is possible to calculate the number of moles, n using the number of molecules and Avogadro's constant.

Identify by placing a tick (✓) in the box next to the correct arrangement of this equation.

Possible equations	Place a tick (✓) here to identify the correct equation
$n = \frac{N_A}{N}$	
$n = \frac{N}{N_A}$	
$n N = N_A$	
$n = N N_A$	

[1 mark]

Question 1c

Define the Avogadro constant.

[2 marks]

Question 1d

The molar mass equation is shown below.

$$n = \frac{m}{M_r}$$

State the meaning of the quantities m and M_r .

[2 marks]

Question 2a

All Ideal gasses obey a relation.

Identify by placing a tick (✓) next to the correct relation.

Possible Ideal Gasses Relations	Place a tick (✓) here to identify the correct relation
$V \propto T$	
$P \propto T$	
$P \propto \frac{1}{V}$	
$pV \propto T$	

[1 mark]

Question 2b

Identify by drawing lines between the name of each gas relation and the name.

Relation	Name of Relation
$\frac{P_1}{T_1} = \frac{P_2}{T_2}$	Boyle's Law
$\frac{V_1}{T_1} = \frac{V_2}{T_2}$	Pressure Law
$P_1 V_1 = P_2 V_2$	Charles's Law

[3 marks]

Question 2c

The following sentences describe the behaviour of an ideal gas.

Identify, by drawing a circle around, the correct word from each pair.

(i)

The molecules in a gas move around **randomly** / **in sequence** at high speeds

[1]

(ii)

The temperature of the gas is related to the **space between** / **average speed** of the particles

[1]

(iii)

Force is the rate of change of **momentum** / **pressure**

[1]

(iv)

Pressure is **increased** / **decreased** if the container of the gas is reduced when the temperature remains constant.

[1]

[4 marks]

Question 2d

An ideal gas is in a container of volume 0.005 m^3 . The gas is at a temperature of 10°C and a pressure of $4 \times 10^5 \text{ Pa}$.

Calculate the pressure of the ideal gas in the same container when it is heated to 30°C .

[4 marks]

Question 3a

There are two ideal gas equations.

Identify by placing a tick (✓) next to the two correct equations.

Possible Ideal Gas Equations	Place a tick (✓) here to indicate the ideal gas equations
$pV = nRT$	
$P \propto \frac{1}{V}$	
$pV = nkT$	
$n = \frac{N}{N_A}$	

[2 marks]

Question 3b

Define an ideal gas.

[2 marks]

Question 3c

This question is about the Boltzmann constant.

(i)

State the equation that defines the Boltzmann constant in terms of the molar gas constant and Avogadro's constant

[1]

(ii)

Use the data booklet to obtain a value for the Boltzmann constant.

[1]

[2 marks]

Question 3d

A cylinder used for scuba diving contains an ideal gas with a volume of $2 \times 10^3 \text{ m}^3$. The gas is at a temperature of 10°C and a pressure of $4 \times 10^2 \text{ Pa}$.

Calculate the amount of gas in the cylinder, in moles.

[3 marks]

Model Answers: Easy

1a

(a) The mole is defined as:

- The SI base unit of an amount of substance; [1 mark]
- It is the amount containing as many particles as there are atoms in 12 g of carbon-12; [1 mark]

[Total: 2 marks]

1b

(b) A table that should score 1 mark would look like this:

Possible equations	Place a tick (✓) here to identify the correct equation
$n = \frac{N_A}{N}$	
$n = \frac{N}{N_A}$	✓ [1 mark]
$n N = N_A$	
$n = N N_A$	

- One mark for the correct equation identified

[Total: 1 mark]

1c

(c) The Avogadro Constant is:

- The number of atoms of carbon-12 in 12 g of carbon-12; [1 mark]

- Equal to $6.02 \times 10^{23} \text{ mol}^{-1}$; [1 mark]

[Total: 2 marks]

1d

(d) The meaning of the quantities are:

- m = the mass of the substance; [1 mark]
- M_r = the molar mass of the substance; [1 mark]

[Total: 2 marks]

2a

(a) A table that should score 1 mark would look like this:

Possible Ideal Gasses Relations	Place a tick (✓) here to identify the correct relation
$V \propto T$	
$P \propto T$	
$P \propto \frac{1}{V}$	
$pV \propto T$	✓ [1 mark]

- One mark for the correct relation ticked

[Total: 1 mark]

There is only **one relation** that identifies an **ideal gas**. The other relations are all gas laws:

- Boyles Law
- Charles's Law
- Pressure Law

2b

(b) An answer that would score 3 marks should look like this:

Relation	Name of Relation
$\frac{P_1}{T_1} = \frac{P_2}{T_2}$	Boyle's Law
$\frac{V_1}{T_1} = \frac{V_2}{T_2}$	Pressure Law
$P_1 V_1 = P_2 V_2$	Charles's Law

- One mark for each correct line drawn

[Total: 3 marks]

2c

(c) The correct sentences are as follows:

- The molecules in a gas move around **randomly** at high speeds; [1 mark]
- The temperature of the gas is related to the **average speed** of the particles; [1 mark]
- Force is the rate of change of **momentum**; [1 mark]
- Pressure is **increased** if the container of the gas is reduced when the temperature remains constant; [1 mark]

(d) Calculate the pressure of the ideal gas in the same container when it is heated to 30 °C:

- Initial temperature, $T_1 = 10^\circ\text{C}$
- Initial pressure, $P_1 = 4 \times 10^5 \text{ Pa}$
- Final temperature, $T_2 = 30^\circ\text{C}$

- Pressure Law is required

- $\frac{P_1}{P_2} = \frac{T_1}{T_2}$ [1 mark]

- $P_2 = \frac{P_1 \times T_2}{T_1}$ [1 mark]

- $P_2 = \frac{(4 \times 10^5) \times 30}{10}$ [1 mark]
- $P_2 = 1.2 \times 10^6 \text{ Pa}$ [1 mark]

(a) A table that would score 2 marks should look like this:

30 °C:

List the known quantities:

- Initial temperature, $T_1 = 10\text{ °C}$
- Initial pressure, $P_1 = 4 \times 10^5\text{ Pa}$
- Final temperature, $T_2 = 30\text{ °C}$

Determine the correct gas law required to calculate the final pressure, P_2 :

- Pressure Law is required
- $\frac{P_1}{P_2} = \frac{T_1}{T_2}$ [1 mark]

Rearrange the equation to make P_2 the subject:

- $P_2 = \frac{P_1 \times T_2}{T_1}$ [1 mark]

Substitute the known quantities and calculate P_2 :

- $P_2 = \frac{(4 \times 10^5) \times 30}{10}$ [1 mark]
- $P_2 = 1.2 \times 10^6\text{ Pa}$ [1 mark]

[Total: 4 marks]

3a

(a) A table that would score 2 marks should look like this:

Possible Ideal Gas Equations	Place a tick (✓) here to indicate the ideal gas equations
$pV = nRT$	✓ [1 mark]

$P \propto \frac{1}{V}$	
$pV = nkT$	✓ [1 mark]
$n = \frac{N}{N_A}$	

- One mark for each correct tick

[Total: 2 marks]

There are only **two equations** used for ideal gases.

- $p \propto \frac{1}{V}$ is Boyles Law
- $n = \frac{N}{N_A}$ enables us to calculate the number of moles

3b

(b) The definition of an ideal gas is:

- A gas which obeys the ideal gas equation; [1 mark]
- At all pressures, temperatures and volumes; [1 mark]

[Total: 2 marks]

3c

(c)

(i) The equation that defines the Boltzmann constant in terms of the molar gas constant and Avogadro's constant is:

- $k = \frac{R}{N_A}$ [1 mark]

(ii) A value for the Boltzmann constant is:

- $k = 1.38 \times 10^{-23} \text{ J K}^{-1}$ [1 mark]

[Total: 2 marks]

3d

(d) Calculate the amount of gas in the cylinder, in moles:

List the known quantities:

- Volume, $V = 2.00 \times 10^3 \text{ m}^3$
- Temperature, $T = 10.0^\circ\text{C}$
- Pressure, $p = 4.00 \times 10^2 \text{ Pa}$
- Molar gas constant, $R = 8.31 \text{ m}^2 \text{ kg s}^{-2} \text{ K}^{-1} \text{ mol}^{-1}$

Use the ideal gas equation in terms of number of moles n :

- $pV = nRT$

- $n = \frac{pV}{RT}$ [1 mark]

- $n = \frac{(4 \times 10^2) \times (2 \times 10^3)}{8.31 \times 10}$ [1 mark]

- $n = 9627 = 9630 \text{ (3 s.f.)}$ [1 mark]

[Total: 3 marks]

All of the quantities in the question, are written to **3 s.f.** and your final answer should be too.

